

Example 6.2

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

$$\Rightarrow X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} = \varepsilon_t$$

$$\rho(h) - \phi_1 \rho(h-1) - \phi_2 \rho(h-2) = 0$$

Yule-Walker eq'ns (Lecture 4)

$$\rho(1) = \frac{\phi_1}{1 - \phi_2}$$

$$\rho(2) = \frac{\phi_1^2}{1 - \phi_2} + \phi_2$$

For the PACF:

$$\rho(o, 1) = \rho(1) = \frac{\phi_1}{1 - \phi_2}$$

$$\rho(o, 2|1) = \phi_2$$

$$= \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

$$= \frac{\left(\frac{\phi_1^2}{1 - \phi_2} + \phi_2 \right) - \frac{\phi_1^2}{(1 - \phi_2)^2}}{1 - \frac{\phi_1^2}{(1 - \phi_2)^2}}$$

$$= \frac{\frac{\phi_1^2 + \phi_2(1-\phi_2)}{1-\phi_2} - \frac{\phi_1^2}{(1-\phi_2)^2}}{1 - \frac{\phi_1^2}{(1-\phi_2)^2}}$$

$$= \frac{\phi_1^2(1-\phi_2) + \phi_2(1-\phi_2)^2 - \phi_1^2}{(1-\phi_2)^2 - \phi_1^2}$$

$$= \frac{\phi_2(1-\phi_2)^2 - \phi_1^2\phi_2}{(1-\phi_2)^2 - \phi_1^2}$$

$$= \frac{\phi_2[(1-\phi_2)^2 - \phi_1^2]}{(1-\phi_2)^2 - \phi_1^2}$$

$$= \phi_2.$$

$$e(o, m \mid 1, \dots, m-1) = 0 \quad m > 2.$$